Machine Learning Based QSAR Modeling for Kinase Drug Discovery

Model Development Validation Interpretability and Applicability Domain Assessment

Abstract

This report evaluates the existing QSAR-X repository without replacing any candidate models. It recomputes predictions from 21 registered model artifacts on their saved scaffold-held-out test partitions. Mean internal scaffold-test R2 was 0.627 and mean RMSE was 0.717 pIC50. These are regression metrics, not biological or clinical accuracy. Strict external quality gates passed for 0 of 21 models.

Current scientific status: target-specific research screening models. No registered model is represented as commercially or prospectively validated.

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1 Project audit and provenance

The repository contains ChEMBL IC50 collection and curation scripts, RDKit structural processing, versioned model files, split assignments, cross-validation comparisons, external evidence reports, a FastAPI prediction service, and a React dashboard. The 20 target-specific models use curated IC50 records converted to pIC50 = 9 - log10(IC50 in nM). The additional JAK candidate uses a user-provided pChEMBL dataset and is retained as a pooled family endpoint.

Audit item	Result
Registered models	21: 20 ChEMBL target-specific and 1 pooled JAK
Molecular representation	RDKit Morgan fingerprint, radius 2, 2,048 bits
Validation partition	Saved scaffold-held-out internal test
External gates passed	0

2 Data quality control

The evaluator checks clean inputs for missing pIC50, duplicate canonical structures, RDKit parse failures, and exact canonical-structure leakage between development and scaffold-test sets. Model SHA256 checksums are retained in the machine-readable audit.

Gene	Clean n	Missing	Duplicates	Dev-test overlap
ABL1	2151	0	0	0
AKT1	3455	0	0	0
BRAF	5384	0	0	0
CDK2	2115	0	0	0
EGFR	10834	0	0	0
ERBB2	2522	0	0	0
FGFR1	3596	0	0	0
FLT3	3766	0	0	0
IGF1R	2324	0	0	0
JAK	10070	0	0	0
JAK2	10405	0	0	0
KDR	9762	0	0	0
KIT	1946	0	0	0
MAP2K1	866	0	0	0
MAPK14	4744	0	0	0
MET	4193	0	0	0
MTOR	4354	0	0	0
PDGFRA	632	0	0	0
PIK3CA	6465	0	0	0
RET	3980	0	0	0
SRC	3601	0	0	0

3 Internal scaffold test evaluation

Existing candidate algorithms were selected using training-only scaffold-grouped CV. For this report, every saved model was loaded and used to recompute its saved scaffold-test predictions. R2, RMSE, MAE, median absolute error, explained variance, Pearson r, Spearman rho, residuals, and bootstrap intervals are written as structured output. The values below are internal validation only.

Gene	Model	n	R2	RMSE	MAE	Spearman
ABL1	SVR	402	0.798	0.683	0.512	0.874
AKT1	SVR	619	0.668	0.731	0.566	0.804
BRAF	SVR	1265	0.720	0.656	0.483	0.855
CDK2	SVR	615	0.569	0.843	0.638	0.758
EGFR	ExtraTrees	2368	0.552	0.850	0.635	0.733
ERBB2	ExtraTrees	474	0.629	0.741	0.571	0.801
FGFR1	SVR	610	0.695	0.657	0.505	0.838
FLT3	SVR	691	0.608	0.783	0.607	0.774
IGF1R	SVR	467	0.769	0.593	0.410	0.872
JAK	ExtraTrees	1864	0.657	0.703	0.523	0.804
JAK2	ExtraTrees	2017	0.716	0.696	0.513	0.827
KDR	ExtraTrees	1909	0.541	0.759	0.580	0.744
KIT	ExtraTrees	348	0.420	0.750	0.570	0.630
MAP2K1	SVR	212	0.419	0.767	0.613	0.539
MAPK14	SVR	1039	0.546	0.720	0.550	0.717
MET	SVR	784	0.629	0.647	0.490	0.809
MTOR	SVR	786	0.747	0.603	0.457	0.853
PDGFRA	SVR	104	0.409	0.798	0.615	0.557
PIK3CA	SVR	1279	0.715	0.615	0.466	0.836
RET	SVR	869	0.616	0.612	0.443	0.751
SRC	ExtraTrees	719	0.733	0.841	0.610	0.886

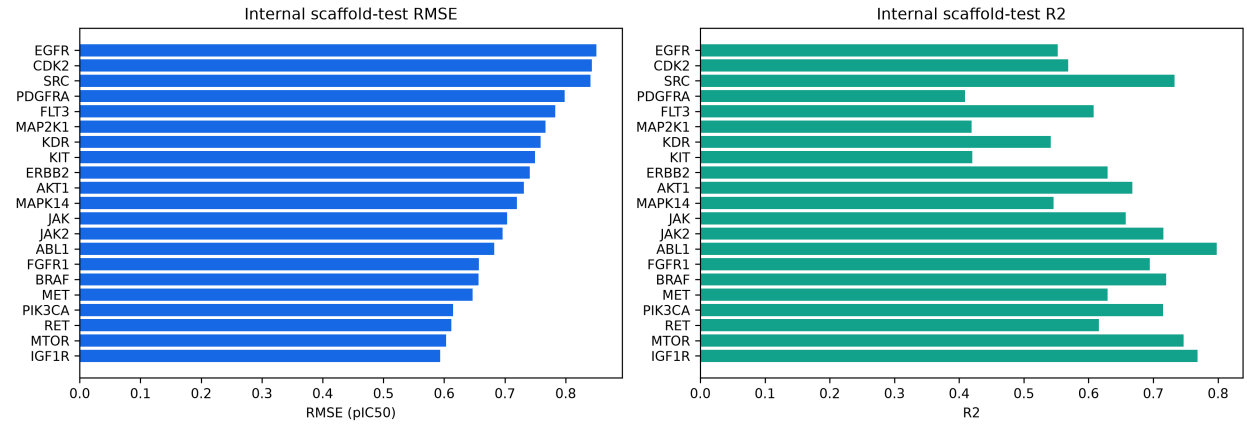


Figure 1. Recomputed internal scaffold-test performance across registered candidates.

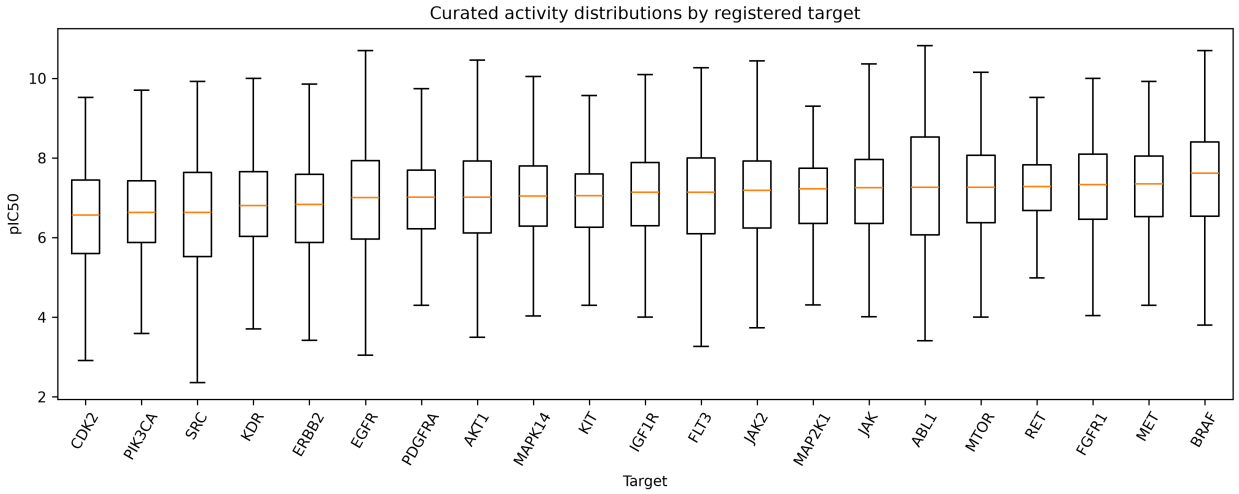


Figure 2. Curated pIC50 distributions by registered target.

4 JAK pooled model evaluation

The pooled JAK candidate was recomputed on 1864 scaffold-test compounds: R2 0.657, RMSE 0.703, MAE 0.523, Spearman rho 0.804. It predicts a pooled pChEMBL endpoint and cannot support JAK1/JAK2/JAK3 isoform-selectivity claims because its source has no per-row isoform label or assay metadata.

Split	R2	RMSE	MAE
Recorded scaffold-held-out	0.657	0.703	0.523

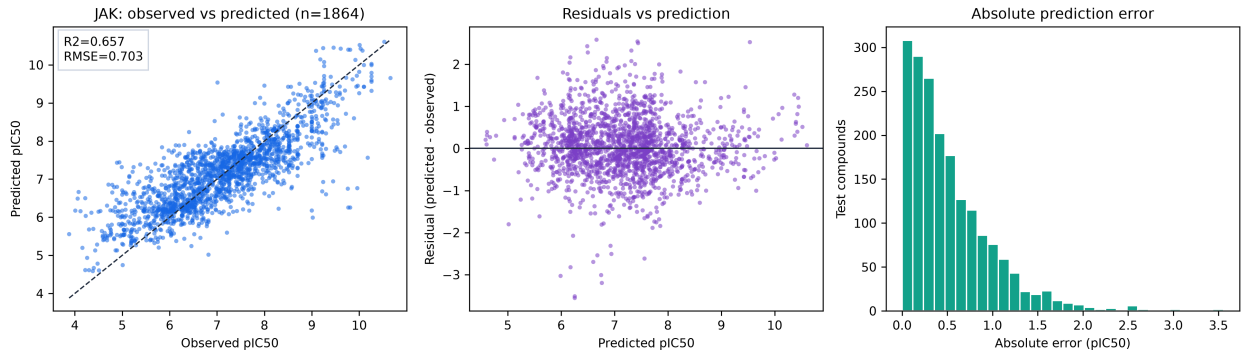


Figure 3. JAK observed-versus-predicted, residual, and absolute-error diagnostics.

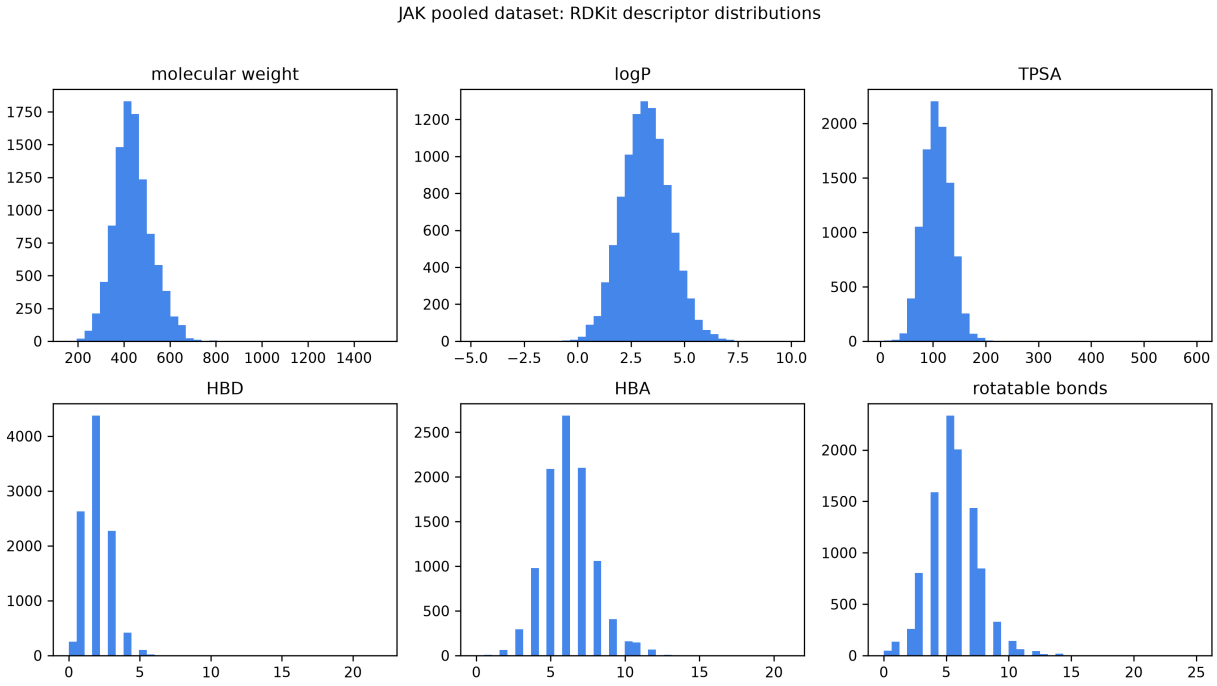


Figure 4. RDKit descriptor distributions for pooled JAK structures.

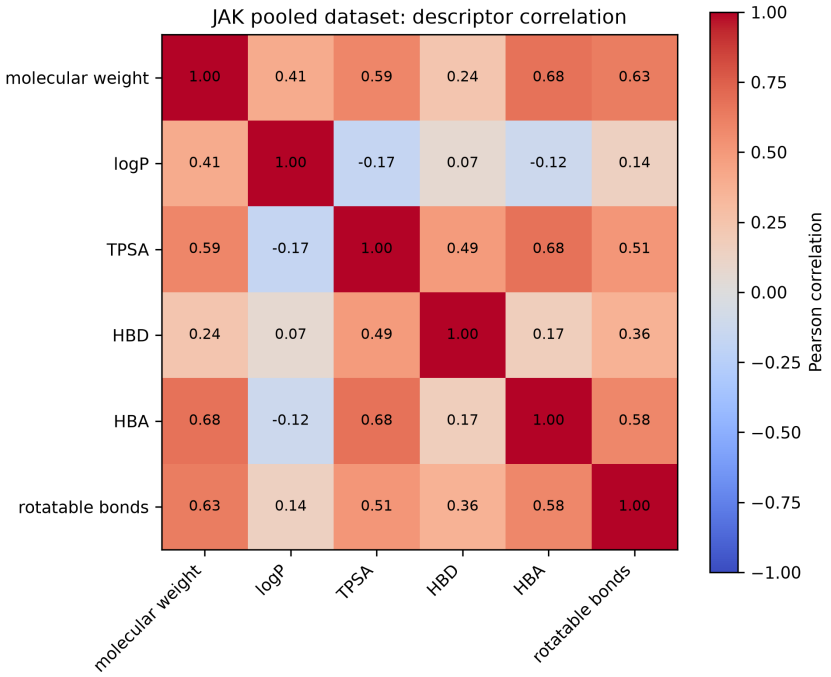


Figure 5. Descriptor correlation heatmap for pooled JAK structures.

5 Applicability domain and interpretability

The JAK applicability-domain file calculates maximum Morgan Tanimoto similarity to the development chemistry for a reproducible 25-compound scaffold-test sample. The sample 5th-percentile threshold is 0.627. This is an empirical internal alert threshold and is not a calibrated probability of correctness. A focused held-out permutation analysis evaluates the 20 highest native-importance fingerprint bits on a fixed 100-compound scaffold-test sample. It is a local sensitivity diagnostic, not a causal potency mechanism or stability analysis.

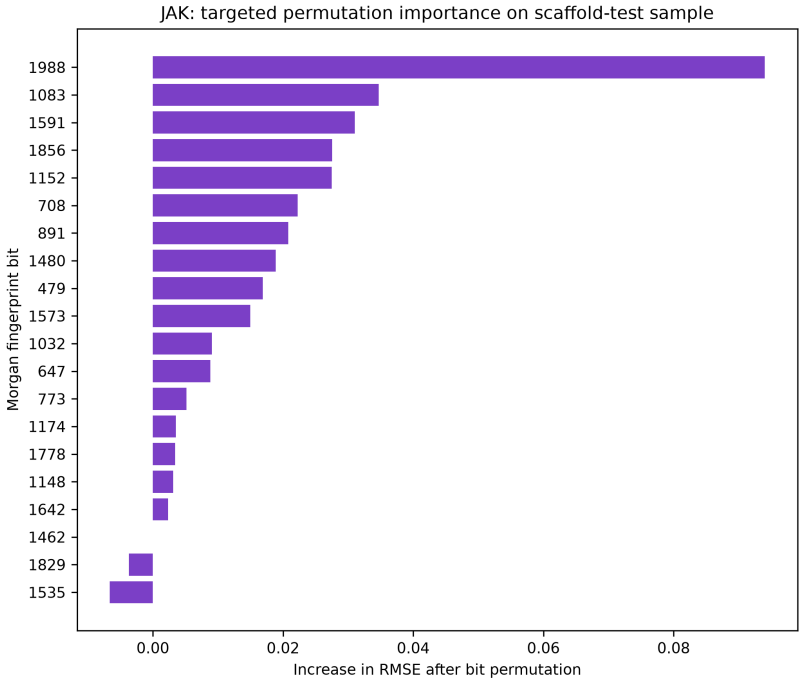


Figure 6. Targeted held-out permutation importance for high native-importance fingerprint bits.

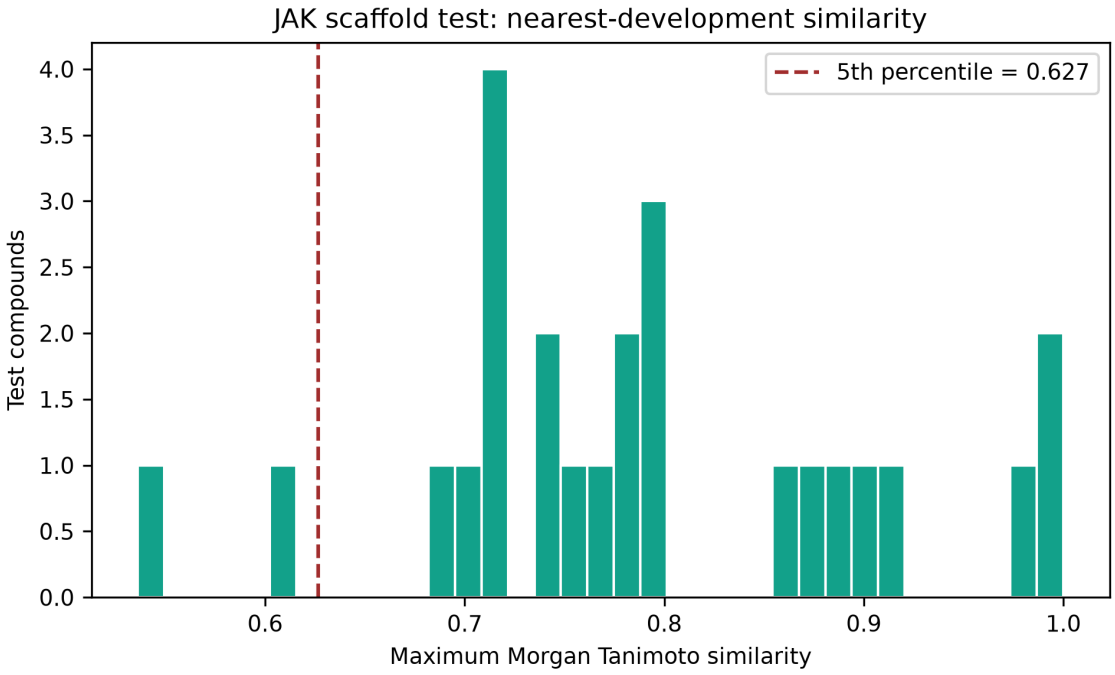


Figure 7. JAK nearest-development similarity distribution for the evaluated sample.

6 External validation and limitations

External validation requires compatible endpoint and units, exact relation, assay context, provenance, structural and publication independence, sufficient sample size, and predefined performance thresholds. The project keeps exploratory external-source checks, but none pass every gate.

Gene	Source	n	RMSE	Spearman	Pass
ABL1	BindingDB REST	139	1.146	0.556	False
AKT1	BindingDB REST	91	0.792	0.608	False
BRAF	BindingDB REST	203	1.228	0.434	False
CDK2	BindingDB REST	741	1.040	0.489	False
EGFR	BindingDB REST	770	1.665	0.366	False
ERBB2	BindingDB REST	1497	1.404	0.215	False
FGFR1	BindingDB REST	103	0.749	0.534	False
FLT3	BindingDB REST	52	0.861	0.723	False
IGF1R	BindingDB REST	33	0.727	0.461	False
JAK2	PubChem BioAssay AID 1699	10	0.333	0.673	False
KDR	BindingDB REST	160	0.829	0.653	False
KIT	BindingDB REST	128	1.010	-0.175	False
MAP2K1	BindingDB REST	63	0.751	0.606	False
MAPK14	BindingDB REST	245	1.148	0.435	False
MET	BindingDB REST	83	0.813	0.820	False
MTOR	BindingDB REST	28	0.707	0.722	False
PDGFRA	BindingDB REST	71	1.073	-0.084	False
PIK3CA	BindingDB REST	566	1.202	0.396	False
RET	BindingDB REST	95	1.413	0.278	False
SRC	BindingDB REST	62	0.638	0.888	False

The models rank candidate chemistry for research. They do not predict clinical efficacy, safety, or an experimental measurement for every arbitrary protein or compound. Internal scaffold testing is not independent external validation. The JAK input’s pChEMBL source cannot be called a verified IC50 dataset.

7 Reproducibility and references

Seed 42; Python 3.14.2; RDKit 2026.03.6; scikit-learn 1.9.1; NumPy 2.5.3; pandas 3.0.6; SciPy 1.18.1. Recomputed predictions, residuals, figures, audit, validation inventory, model checksums, and manifests are stored under results/.

OECD. Guidance Document on the Validation of (Quantitative) Structure-Activity Relationship [(Q)SAR] Models. OECD Series No. 69, 2014. https://doi.org/10.1787/9789264085442-en

RDKit documentation. The RDKit Book: Morgan and Feature Morgan fingerprints. https://www.rdkit.org/docs/RDKit_Book.html

Rogers D, Hahn M. Extended-connectivity fingerprints. J Chem Inf Model. 2010;50:742-754. https://doi.org/10.1021/ci100050t